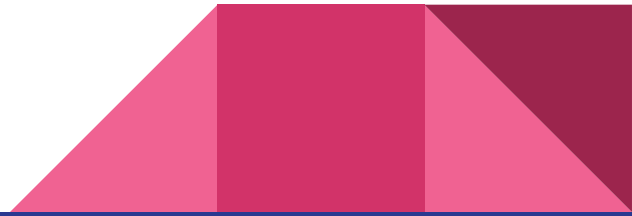
Assignment I

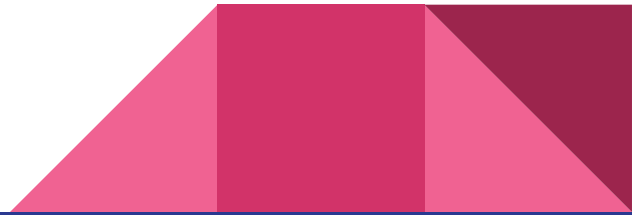
Exercise

1. Area of a rectangle: Prompt the user to **enter the length and width** of the rectangle as integers. Calculate the area using the formula **area = length * width** and print the result.
2. Average of three numbers: Prompt the user to enter three numbers as integers. Calculate the average of 3 numbers. Display the average
3. Swaps the values of two variables: Prompt the user to enter two numbers as integers. Print the values before and after the swap



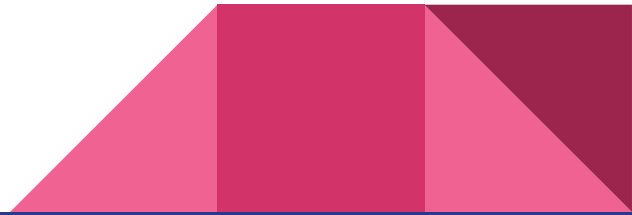
Exercise

1. Book Discount: Suppose the cover price of a book is $24.95, and a bookstore offers a 40% discount. Calculate and display the final price of the book (rounded to 2 decimal places).
2. Time Conversion: Calculate how many seconds are there in 42 minutes 42 seconds.
3. Distance Conversion: Calculate how many miles are in 10 kilometers. (Hint: There are 1.61 kilometers in 1 mile.).
4. Average Speed: If you run a 10-kilometer race in 42 minutes and 42 seconds, what is your average speed in kilometers per hour (km/h)?



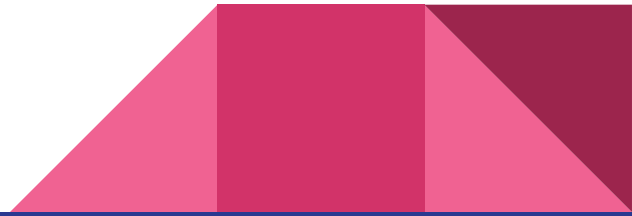
Exercise

1. Voting Eligibility: Write a program that prompts the user to enter their age and checks if they are eligible to vote (18 years or older).
2. Store Discount Calculator: Write a program that prompts a cashier to enter a purchase amount and calculates the final price based on the following discount tiers:
    a. 10% discount if above 100$
    b. 20% discount if above 500$
    c. 50% discount if above 1000$



Exercise

1. Electricity Bill Calculator: Write a program that prompts the user for their total electricity consumption (in kWh) and calculates the total bill in Riel using the rates below:
    a. From 0 to 50KW, price 1Kwh = 500 Riel
    b. From 50 to 100KW, price 1KW = 620Riel
    c. 100KW+, price 1kw = 750 Riel
2. Net Salary Calculator: Write a program that takes a user's gross monthly salary (in Riel) and calculates their net salary after deducting income tax based on the following brackets:
    a. 0 to 1,500,000Riel, tax is 0%
    b. 1,500,001 to 2,000,000, tax is 5%
    c. 2,000,001 to 8,500,000, tax is 10%
    d. 8,500,001 to 12,500,000, tax is 15%
    e. Over 12,500,000, tax is 20%



Exercise

1. Telco Carrier Detection: Write a program that repeatedly prompts the user to enter a phone number until it meets all of the following criteria:
    i. User must input number only
    ii. User must input number 9 to 10 number
    iii. User must input one of carrier prefix (cellcard, smart, and metfone)
    b. Print message “Your number carrier is Cellcard/Smart/Metfone”
2. Special Account Number Purchase: Write a program that continuously prompts the user to enter their preferred account number
    a. Format the number as a 6-digit string padded with leading zeros.
    b. Keep track of already-purchased numbers
    c. Example of output
        i. if user input 1, then print “You have purchased 000001 !”
        ii. If user input 8550, then print “You have purchased 008550 !”
        iii. If user input 8550 again, then print ”This number has been purchased”